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To cite this article: Esa M. Rantanen & Christopher D. Wickens (2012) Conflict Resolution Maneuvers in Air Traffic Control: Investigation of Operational Data, *The International Journal of Aviation Psychology*, 22:3, 266-281, DOI: [10.1080/10508414.2012.691048](https://doi.org/10.1080/10508414.2012.691048)

To link to this article: <https://doi.org/10.1080/10508414.2012.691048>



Published online: 02 Jul 2012.



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Conflict Resolution Maneuvers in Air Traffic Control: Investigation of Operational Data

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Little is known about air traffic controllers' preferred maneuvers to resolve conflicts between pairs of aircraft, despite the topic's importance of to the planned NextGen system. We analyzed 256 cases of conflict resolution maneuvering from radar track data from 5 U.S. air traffic control centers. Our results show the influences of conflict geometries on controllers' maneuver choices.

The NextGen air traffic control (ATC) and air traffic management (ATM) will bring about fundamental changes in the roles, tasks, and task elements of air traffic controllers and pilots operating the future system. Perhaps the most dramatic of these changes will be shared conflict detection, avoidance, and resolution among controllers, pilots, and automated, computational algorithms. Despite much past experience on this concept both in ATC (conflict alert [CA]; Nolan, 1998) and in cockpits (Traffic Alert and Collision Avoidance System [TCAS]; Bliss, 2003; Chappell, 1990), procedures for shared conflict avoidance and resolution are still largely on the drawing board. Before these procedures are finalized and implemented, it is critical to understand their implications in the context of current entrenched conflict resolution techniques and rules of the air.

PAST RESEARCH

There is a relatively large body of research literature on conflict resolution maneuver conventions among pilots (e.g., Alexander, Merwin, & Wickens, 2005; Smith, Ellis, & Lee, 1984; Thomas & Rantanen, 2006; Thomas & Wickens, 2008) due

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to the emergence of airborne collision detection systems (e.g., TCAS and cockpit display of traffic information [CDTI]) in the past couple of decades. Past research on CDTI has revealed that pilots tend to prefer vertical maneuvers over lateral ones, reflecting perhaps the greater expediency and reduced complexity of such maneuvering (Thomas & Wickens, 2008). However, the data are somewhat ambiguous regarding the extent to which the particular geometry of a conflict dictates the direction of a maneuver. These systems and the maneuvers initiated by them or by pilots are moreover quite different from maneuvers instructed by air traffic controllers.

Less data are available on air traffic controllers' conflict avoidance maneuver preferences. The ATC task environment and controllers' tasks are also markedly different from those of pilots. Pilots are primarily concerned about their own aircraft and their attention extends little beyond those other aircraft that pose an immediate or near immediate threat of loss of separation, whereas air traffic controllers must concern themselves with the "big picture" and traffic flows in addition to individual aircraft pairs. Moreover, controllers are always responsible for a much larger number of aircraft than any pilot in any situation, and effective management of their own workload is critical to their performance. Among the few laboratory studies on controller maneuver preferences, Rantanen, Yang, and Yin (2006) used a low-fidelity simulator with a simplified airspace and traffic patterns. The results showed strong preference for vertical separation (i.e., instructing planes to climb or descend, or to remain at an intermediate altitude) over vectoring (lateral maneuvers). This is consistent with Rantanen and Nunes's (2005) findings of hierarchical conflict detection strategies of controllers, where vertical separation was always considered first.

Little information is available about the ways controllers are taught to resolve conflicts. The Federal Aviation Administration (FAA) Academy lesson plan (FAA, 2008) provides definitions and application guidelines for each type of standard separation, but no particulars whatsoever about conflict geometries or how to resolve various conflicts in an effective and expeditious manner. We assume that such know-how is transferred to novice controllers by their instructors during their training in the Academy or on-the-job training in their facilities. The variability in conflict resolution techniques learned and developed by individual controllers during training and throughout their careers is impossible to estimate, but observation of controllers working with live traffic or analysis of relevant data collected from operational settings might offer insights into common practices.

THEORETICAL FACTORS INFLUENCING MANEUVER CHOICE

In the absence of any clear guidance from controller training or doctrine regarding appropriate maneuvers, it is possible to adopt the procedure used by Wickens,

Helleberg, and Xu (2002) and Thomas and Wickens (2008) and define the factors that should combine to influence the *expected utility* (EU) of the controller's decision to instruct one type of maneuver over another. Accordingly, we can identify three main influences: (a) expediency, that is, maneuvers that can resolve a conflict more rapidly are preferred over time-consuming ones, (b) preservation of airspace structure or maneuvers that are least disruptive to the overall flow are preferred, and (c) visualization, or maneuvers whose implementation can be more readily perceived on the controller's traffic display are preferred.

To elaborate, three further factors influence expediency: (a) control order, (b) gravity, and (c) lateral geometry. In particular lateral maneuvers, requiring aircraft turns, are of higher order (second) than vertical ones (first order). Fundamental laws of control theory dictate the more rapid execution of lower order (vertical) control variables (Wickens, 2003). In a similar manner, any vertical maneuvers that exploit gravity (descending or leveling off from a climb) will be more rapid than those that oppose gravity (climbing, or leveling off from a descent). Reversals of vertical trajectories (instructing a climbing aircraft to descend or descending aircraft to climb) are especially to be avoided due to the difficulty of flying such maneuvers and the time their execution will require. Finally, when two aircraft are on crossing headings at the same altitude, any turning maneuver will be slower in resolving the conflict than when they are flying on parallel tracks (whether opposite headings, or on a same heading and overtaking).

With respect to preservation of the airspace structure and traffic flows, lateral movements tend to be more disruptive than vertical ones. Vertical separation between aircraft is the easiest one to implement and requires the least amount of attention and vigilance from controllers (cf. Rantanen & Nunes, 2005). Two aircraft at different altitudes will not be in conflict no matter what they do as long as they maintain their levels. Turns, on the other hand, although resolving one conflict, will place aircraft in potential conflicts with several other aircraft at the same altitude. Furthermore, ATC sectors in busy airspaces are commonly very narrow and corridor-like, with little room for lateral maneuvers and deviations from the nominal traffic flow, requiring constant vigilance to return a vectored aircraft to it before violating sector boundaries. Aircraft in vertical transitions (i.e., climbing or descending) can also be quite troublesome for the same reasons, as they cross levels of several other aircraft requiring a lateral or longitudinal separation from them. Level-offs at intermediate altitudes are therefore often an easier solution to conflicts than vectoring, which would in this case involve deviations from two constants (altitude and planned or standard horizontal trajectory) simultaneously.

Lastly, two factors influence visualization of conflicts and their resolutions on controllers' plan view displays (PVDs). Lateral maneuvers are easier to visualize, as their effects are graphically represented on the PVD, whereas vertical maneuvers require interpretation of numerical altitude data in the aircraft data blocks. Level-offs are also easier to visualize than climbs or descents, because the former

TABLE 1
The Hypothesized Expected Utilities (EUs) of Different Conflict Resolution Maneuvers, After Wickens, Helleberg, and Xu (2002)

	<i>1.1. Exp- Order</i>	<i>1.2. Exp- Gravity</i>	<i>1.3. Exp-Lat. Geo.</i>	<i>2. Airspace and Flow</i>	<i>3. Visual</i>	<i>EU Overall</i>	<i>EU Crossing</i>
Climb	1		1	1		2	3
Descend	1	1	1	1		3	4
Climb-level-off	1	1	1	1	1	4	5
Descend-level-off	1		1	1	1	3	4
Turn					1	1	1

will maintain the altitude at a constant value, and the latter will require the controller to attend a continually changing digitally represented altitude to determine or monitor rate of climb or rate of descent.

Collectively then, these factors allow the assignment of a “utility score” to each of the five major classes of maneuver types, as shown in Table 1. In Table 1, the five factors hypothesized to influence maneuver choice are listed across the first five columns, and the five maneuver categories are listed down the rows. Each maneuver that satisfies a criterion (factor) is assigned a 1 value utility in the appropriate cell. If the cells are summed across the columns, a total EU score is presented in the sixth column. Finally, the last column amends that EU score contingent on a crossing horizontal track. Although these sums provide measures of EU only on an ordinal scale, they might nevertheless be quite useful in predicting controllers’ choice of maneuvers to resolve conflicts.

It might be noted that the preceding approach is also represented in many computational conflict resolution algorithms, which often include a prominent cost factor. It would indeed be interesting to compare the theoretical factors affecting human controllers’ conflict resolution maneuvers as well as those they actually choose in operational settings to those devised by computational algorithms. This is quite a difficult task, however, and beyond the scope of this article. Computational conflict resolution algorithms tend to be very complex with a myriad of variables, many of which are only vaguely defined in the open literature (e.g., Hu, Prandidi, Nilim, & Sastry, 2002). We can reasonably assume, however, that human controllers intuitively take into account many of the same variables as are included in the computational algorithms, as well as other variables not part of automated conflict resolutions (e.g., visualization), in their everyday work.

PURPOSE OF THE STUDY

In spite of the importance of this topic for aviation safety, and in particular for understanding how a controller’s tendencies can either reinforce or contradict

pilot tendencies, no data appear to exist regarding the actual controller conflict avoidance behavior with live traffic. The intent of this article is to bring insight to this process, using the operational ATC en route data from controllers responding to CAs at five Air Route Traffic Control Centers (ARTCCs; see Wickens et al., 2008, for details). Our focus is on two key aspects of the data: (a) what tendencies do controllers show in terms of instructing lateral versus vertical maneuvers, and, within the latter category, climbs versus descents versus level-offs, and (b) how are these tendencies mediated or influenced by the particular geometry of a conflict. A third aspect of these data pertaining to how controllers' responses are mediated by alert reliability (e.g., false alert rate) is reported in detail by Wickens, Rice, Keller, Hughes, and Hutchins (2009).

Hypotheses

We formulated several hypotheses based on the preceding discussion and the analysis presented in Table 1, for which mean EU scores are derived and shown in parentheses later. Specifically, we hypothesized that (a) vertical maneuvers ($M = 4$) would be preferred over lateral ones ($M = 1$) based on previous research (Rantanen et al., 2006; Thomas & Wickens, 2008), (b) descents and level-offs from climbs ($M = 3.5$) should be preferred over climbs and level-offs from descents because of gravity ($M = 2.5$), (c) level-offs ($M = 4.5$) should be preferred to climbs or descents ($M = 3.5$), (d) turns would be used more with acute and obtuse conflict angles (same and opposite headings) due to their greater efficiency in these cases than with trajectories crossing close to right angles, and (e) controllers would respond earlier in conflicts with opposite or crossing trajectories and large vertical speed differences because they produce higher rates of closure than acute conflict angles and small vertical speed differences. We also viewed conflict variables as continuous (e.g., conflict angle and vertical closure speed), and formulated two additional hypotheses: (f) If resolution maneuvers were contingent on the horizontal conflict geometry, the conflict angles would be clustered according to the contingent maneuver. (g) We also expected different maneuvers to be instructed depending on the rate of vertical convergence, reflecting different levels of urgency in a conflict situation. Finally, because the cases we studied were in response to CAs and thus we had controllers' response times available, we expected (h) to see "distance over speed" bias where CAs at large angles and high vertical convergence rates would delay responses (cf. Xu & Rantanen, 2007).

METHOD

Recently, available operational data have allowed for examination of controllers' conflict resolution maneuvers "in the wild." Wickens et al. (2008; Wickens et al.,

2009) reported how 495 en route controllers' responses are mediated by alert reliability (e.g., false alert rate). The data included a large set of conflict cases from five ARTCCs: Houston (ZHU), Indianapolis (ZID), Salt Lake City (ZLC), Los Angeles (ZLA), and Atlanta (ZTL). These data were originally provided to researchers at the FAA Technical Center (Allendoerfer, Friedman- Berg, & Pai, 2007) for analysis, and to us by the FAA in cooperation with the National Air Traffic Controllers' Association. Data for each aircraft pair in conflict consisted of predicted point of closest approach, time of alert, and the radar tracks and altitudes of the aircraft, allowing for analyses of the actual conflicts as they played out. Alas, no voice transcripts were associated with the data, so inferences of an instructed maneuver had to be made solely from trajectory changes following the CA (see Allendoerfer et al., 2007).

The analyses reported here included 223 cases where a conflict avoidance maneuver could be reliably discerned; 33 cases where both aircraft in a conflict pair maneuvered were included in the analyses as two separate maneuvers, resulting in a total of 256 cases. These cases came from the same five ARTCCs mentioned earlier, ZID ($N = 128$, or 50%), ZTL ($N = 81$, or 31.64%), ZHU ($N = 38$, or 14.84%), ZLC ($N = 6$, or 2.34%), and ZLA ($N = 3$, or 1.17%).

Categorical Data

The data were analyzed as follows: For each case, approximately 6 min worth of actual track data of the two aircraft in conflict were recorded, including the x- and y- (latitude and longitude) and z- (altitude) coordinates sampled every 10 sec. These coordinates were plotted separately for horizontal and vertical trajectories, from where the conflict geometries and maneuvers performed to resolve the conflict could be determined visually. The geometries were classified into three vertical categories (converging vertically, parallel climbs or descents, or both aircraft level) and three horizontal categories according to the FAA course definitions in the ATC handbook JO 7110.65T (FAA, 2010) (same within 45°, crossing 45° to 135°, or opposite, over 135°). Four classes of maneuvers were defined: descend, climb, level-off for aircraft in descent or climb, and turn. In the case of joint maneuvers in both the lateral and vertical axes, the CA was assigned to the category of the maneuver that occurred first (earliest). These data were tabulated and analyzed by chi-square tests for independence.

Continuous Variables

Categorical analysis is a common and undeniably valuable way to examine safety data, but its limitations must be acknowledged. Conflict geometries exhibit enormous variability and any classification system necessarily includes very different situations forcing different maneuver choices into the same categories. The

purpose of these additional analyses was to view conflict geometries as a continuous variable allowing more fine-grained measurement of their characteristics, and the effect these characteristics might have on subsequent conflict resolution maneuvers.

Conflict angle. From the same data described earlier, the exact horizontal conflict angles and differences between vertical speeds of the aircraft in a pair were calculated. The latter measure represents the rate of vertical convergence between aircraft. Exact conflict angles were computed by calculating the slopes of the straight proportions of the aircraft trajectories before a conflict avoidance maneuver. The slopes were given by the equation

$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad (1)$$

where the x , y coordinates were obtained from the actual aircraft track data in the data files. The conflict angle (AOC) was then calculated as the angle between the slopes of the two aircraft in conflict:

$$AOC = \arctan \left| \frac{m_1 - m_2}{1 - m_1 m_2} \right| \quad (2)$$

Note that Equation 2 does not guarantee the correct angle, as the slopes contain no directional information. The resulting angle was therefore visually checked against the plot of the conflict geometry, and in case the actual conflict angle was obtuse, the angle given by Equation 2 was subtracted from 180. The vertical speed difference was calculated simply as the absolute value of first aircraft's vertical speed subtracted from the second aircraft's vertical speed.

We plotted conflict angles against four maneuver classes—climb, descend, level-off (including both reduced climb and descent), and turn—to determine whether these maneuvers were contingent on the horizontal conflict geometry. In a similar manner, vertical speed differences were plotted against maneuver categories to see if the rate of vertical convergence influenced maneuvers to be instructed.

Response time. We calculated controllers' response time to the CA by subtracting CA time from maneuver time in the data files. In some cases this time was negative, meaning that the controller reacted to the conflict before the automated CA (see Wickens et al., 2009). Plotting these response times against conflict angles and rates of vertical convergence, we expected to see later responses to obtuse horizontal conflict angles, which are more difficult to evaluate than conflict angles that are acute. We also expected earlier responses to conflicts with

high vertical convergence rates. On one hand, because all the data were cases where the CA system was triggered, and because CA is a time-based system that takes into account higher convergence rates at large horizontal angles and at high vertical speed differences, CA would have occurred at larger spatial separation in these cases than at smaller angles and vertical speed differences. On the other hand, a “distance over speed” bias (Xu & Rantanen, 2007) would predict later responses to CAs at large angles and high vertical convergence rates. To elaborate, in two experiments, Xu and Rantanen (2007) had pilots observe traffic on CDTI; the pilots assessed conflicts as more severe if planes were closer (but converging more slowly) than farther away (and converging more rapidly), exhibiting a bias to be influenced more heavily by the closer distance than the lower speed information in estimating time to closest point of approach.

RESULTS

Categorical Analyses

Level-offs constituted the most frequent maneuver category (44%), followed by turns (32%), descents (18%), and climbs (5%). Turns occurred much less frequently than vertical trajectory changes (32% vs. 68%), however. This result is consistent with the integrated findings of studies of pilot-initiated conflict avoidance (Thomas & Wickens, 2008). Descents (exploiting gravity) were instructed more than three times as frequently as climbs (opposing gravity; 18% vs. 5%), but modifications to vertical transitions already in progress (i.e., level-offs) were still the most frequent maneuvers (44%). These results clearly support our first and third hypotheses.

The preceding analyses concerned the frequencies in the maneuver categories, independent of the conflict geometry. We also analyzed contingencies between the conflict geometry as perceived by controllers on their display, and the types of maneuvers that were instructed, by chi-square tests for independence. Two contingency tables were created for vertical and lateral geometries and corresponding maneuvers and their combinations. To create the vertical geometry table we used the three vertical conflict geometry classes—vertically converging, both aircraft level, and parallel climbs or descents (i.e., zero or negligible relative vertical speed)—and four maneuver classes (climb, descend, level-off, and turn), for a 3×4 table (Table 2). The results for the vertical geometries were significant, $\chi^2(6, N = 256) = 96.10, p < .001$. This nonindependence is most noticeable in the departure from expectancy for descents, which occurred less than expected on vertically converging, and much more than expected on both-level trajectories.

Similarly, three horizontal geometries (same direction, crossing, and opposite) were analyzed against the aforementioned four maneuver categories in another

TABLE 2
Counts of Different Maneuvers by Vertical Conflict Geometries (Expected
Values in Parentheses)

<i>Vertical Geometry</i>	<i>Climb</i>	<i>Descend</i>	<i>Level-Off</i>	<i>Turn</i>	<i>Total</i>
Converging	5 (8.91)	21 (29.93)	88 (71.95)	49 (52.21)	163
Level	7 (2.13)	25 (7.16)	0 (17.21)	7 (12.49)	42
Parallel	2 (2.95)	1 (9.91)	25 (23.84)	26 (17.30)	54
Total	14	47	113	82	256

TABLE 3
Counts of Different Maneuvers by Horizontal Conflict Geometries (Expected
Values in Parentheses)

<i>Horizontal Geometry</i>	<i>Climb</i>	<i>Descend</i>	<i>Level-Off</i>	<i>Turn</i>	<i>Total</i>
Crossing	12 (11.81)	36 (39.66)	98 (95.34)	73 (69.19)	216
Opposite	0 (0.66)	5 (2.20)	5 (5.30)	2 (3.84)	12
Same	2 (1.53)	6 (5.14)	13 (12.36)	7 (8.97)	28
Total	14	47	113	82	256

TABLE 4
Counts of Different Maneuvers by Combined Horizontal and Vertical Conflict Geometries
(Expected Values in Parentheses)

<i>Horizontal-Vertical Geometry</i>	<i>Climb</i>	<i>Descend</i>	<i>Level-Off</i>	<i>Turn</i>	<i>Total</i>
Crossing-converging	4 (7.55)	12 (25.34)	75 (60.91)	47 (44.20)	138
Crossing-level	6 (1.80)	23 (6.06)	0 (14.57)	4 (10.57)	36
Crossing-parallel	2 (2.46)	1 (8.26)	20 (19.86)	22 (14.41)	45
Same/Opp-converging	1 (1.37)	9 (4.59)	13 (11.04)	2 (8.01)	25
Same/Opp-level	1 (0.33)	2 (1.10)	0 (2.65)	3 (1.92)	6
Same/Opp-parallel	0 (0.49)	0 (1.65)	5 (3.97)	4 (2.88)	9
Total	14	47	113	82	256

3 × 4 table (Table 3). The results were not significant, $\chi^2(6, N = 256) = 6.41, p = .38$, suggesting little or no dependence of maneuver choices on lateral geometry.

Yet another analysis was done on combinations of vertical and horizontal geometries. Combining opposite and same courses into a single category yielded a 6 × 4 table (Table 4). Inclusion of same and opposite courses in the same category was justified by the fact that these represented small (<45°) absolute angles of convergence whereas crossing courses represented large (45°–135°) angles, and to test a hypothesis that fewer turns would be instructed at the latter than the former

angles. Indeed, the chi-square analysis for nonindependence yielded significant results, $\chi^2(15, N = 256) = 115.76, p < .001$. Examination of conflicts where both aircraft were level revealed fewer turns than expected on crossing courses but close to expected frequencies on opposite or same courses. This finding is consistent with the fourth hypothesis.

Analyses of Continuous Variables

Analyses of continuous variables did not bear out our expectations as expressed in our sixth and seventh hypotheses. No discernible patterns could be seen in the plots of horizontal conflict angles or vertical speed differences versus maneuvers. We performed a linear discriminant analysis to classify conflict angles into the four maneuver classes, but only 21% of the angles were correctly classified. Likewise, only 30% of the vertical speed differences were correctly classified in the maneuver groups. This means that conflict angle or vertical speed difference do little to predict the subsequent conflict avoidance maneuver.

Scatterplots of response time to CA versus horizontal conflict angle and vertical speed difference suggested no relationship between these variables (Figure 1). The apparent insensitivity to the urgency of the conflict situation supports the hypothesized "distance over speed" bias. Examining histograms of vertical speed difference and conflict angle by each maneuver did not reveal any particular patterns, with one exception. Histograms of vertical speed difference revealed nearly identical and positively skewed distributions for each maneuver category. Histograms of conflict angles showed mostly uniform distributions for climbs, descents, and level-offs, but a distinctly bimodal distribution for turns, as shown in Figure 2. Turns were thus instructed much more often at small ($<45^\circ$) or large ($>135^\circ$) angles of convergence and less at angles close to 90° (i.e., the crossing geometries of Table 2). These results are consistent with the chi-square analyses reported earlier, providing further support for our fourth hypothesis.

Maneuvering Aircraft State

The state of the aircraft commanded to maneuver (i.e., whether the aircraft was climbing, descending, or in level flight) had a significant effect on the commanded maneuver, $\chi^2(6, N = 256) = 154.05, p < .001$. This statistic is no doubt inflated by the zero frequencies in vertical reversal cells (requesting a climbing plane to descend or descending plane to climb) and the obviously impossible case of leveling-off an aircraft already in level flight (Table 5). There were very few cases where the rate of climb or descent was increased to resolve a conflict (6 and 5, respectively), but as such maneuvers have limited effectiveness, the small numbers are not surprising. There are, however, several noteworthy results in Table 5. First, we should note the much higher than expected frequency of descents than

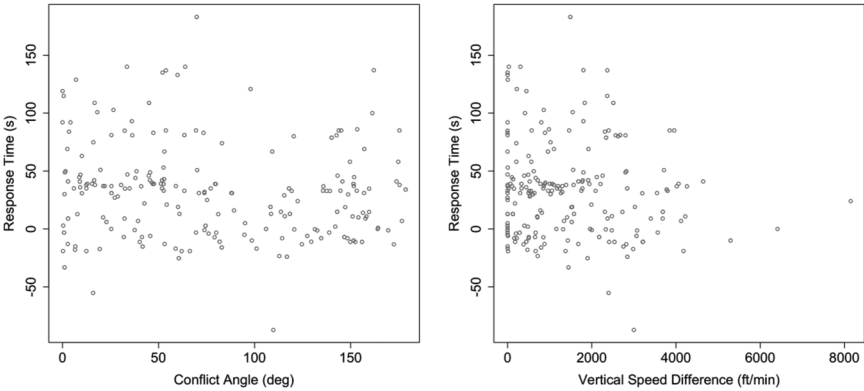


FIGURE 1 Scatterplots of response time to conflict alert (negative reaction time signifies early response) versus horizontal conflict angle (left panel) and vertical speed difference (rate of reduction of vertical separation; right panel). No relationship between these variables is apparent.

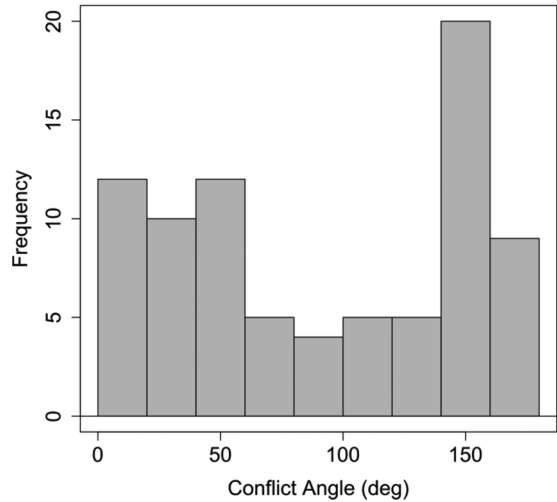


FIGURE 2 Histogram of turning maneuvers at different conflict angles. The bimodal distribution of turns is apparent, indicating that turns are more readily commanded at small (acute) or large (obtuse) convergence angles than at angles close to 90°.

climbs requested from aircraft in level flight. Such a maneuver takes advantage of gravity, supporting our second hypothesis. Level-offs from climbs were also more frequent than from descents (57% vs. 43%). Second, modifications to the vertical trajectory were preferred much more frequently than expected, attesting

TABLE 5
Counts of Different Maneuvers by Maneuvering Aircraft State (Expected
Values in Parentheses)

<i>AC State</i>	<i>Climb</i>	<i>Descend</i>	<i>Level-Off</i>	<i>Turn</i>	<i>Total</i>
Climb	6 (5.69)	0 (19.09)	64 (45.91)	34 (33.31)	104
Descend	0 (4.76)	5 (15.97)	49 (38.40)	33 (27.87)	87
Level-off	8 (3.55)	42 (11.93)	0 (29.69)	15 (20.82)	65
Total	14	47	113	82	256

to the relative ease of such solutions to conflicts, supporting the third hypothesis (see Wickens et al., 2009, for further discussion of anticipatory responses).

DISCUSSION

The two convergent analyses both provide several insights into the extent to which controllers take account of the particular conflict geometry as they formulate their instructions following a CA. Our results are also quite significant in other than a statistical sense because the large sample size provided considerable statistical power to confirm the absence of certain trends. First, critically, the continuous analyses of the data in Figure 1 indicate conclusively that controllers' sense of urgency in their response (as inferred from their response time) is not influenced by either the angle of lateral or vertical closure (the latter expressed as relative vertical velocity). This could indicate the "distance over speed" bias recently observed by Xu and Rantanen (2007). Other factors must be at play to predict the variance in response time, at least for those responses associated with the CA. The timing of earlier issued instructions can indeed be influenced by geometry, but we did not have access to sufficient amount of geometry data for the cases where actions were taken before CA.

Second, both analyses converged to reveal a marked avoidance of issuing turn instructions when the two aircraft were on a crossing pattern. This tendency can be explained by smaller turns necessary to achieve adequate separation at the same or opposite courses than at crossing courses.

Third, the categorical, but not the continuous, vertical geometry analysis revealed that controllers avoided a "descent" instruction when two aircraft were in a vertically converging path. This tendency also makes sense because in such geometry the lower aircraft will likely be climbing and an instructed descent to this aircraft will involve a total reversal of vertical geometry; alternatively, the upper aircraft will likely be descending, and instructing a faster descent to an already descending aircraft could endanger the flight; such a maneuver might be restricted to the more urgent TCAS advisories. The reason why this tendency was manifest in the categorical but not the continuous analysis might reflect the fact

that controllers often think about conflicts in categorical terms, indeed “lumping” all lateral angles into the three preceding categories in terms of triggering instructions, and making similar categorical judgments about vertical behavior. Hence the categorical analyses could be more “tuned” to the mental model of the controller.

It is particularly important to note that both analyses revealed, at most, only very mild contingency effects. This indicates that, at least with the mild urgency of the CA, controllers probably do not delay to “optimize” the choice of a maneuver, but rather, probably use more predetermined rules to select (favoring level-offs, then turns, then descents and then climbs; see Table 4) and then presumably examine if the particular geometry makes the chosen maneuver unsafe (cf. Rantanen & Nunes, 2005). Such rule-based conflict resolutions would create the mild contingencies that were revealed by both forms of analysis.

In terms of the overall maneuver tendencies, independent of the contingent conflict geometry, Table 1 presented a multiattribute choice model of the utility of the different maneuver categories; the chi-square analysis of the data in Table 2 indicated clearly significant differences in the frequency of those different maneuvers. Figure 3 presents a scatter plot showing the relationship between predicted (Table 1) and obtained (Table 2) maneuver frequencies.

The data in Figure 3 reveal two clear findings. First, the four vertical maneuvers that lie to the lower right of the scatter plot show a near perfect agreement between predicted and obtained frequencies, $R = 0.95$, R^2 adjusted = 0.91, $F(2, 4) = 30.12$, $p = .032$. For example, the model’s prediction of highest popularity of the level-off from climb, which exploits gravity, is easy to visualize, preserves lateral airspace structure, and is rapidly executed, is borne out by the highest frequency of observed maneuvers. The two midpoints represent the trade-off between the descent and the level-off descent—the former being difficult to visualize, but the latter working to oppose gravity—and the lowest point, the climb, representing both the opposition to gravity and the greater visualization effort. Indeed the relative positioning of these four points perfectly accounts for the equal influence of the factors in the second (gravity) and third hypotheses (visualization).

The second point graphically illustrated by Figure 3 is that turns are underpredicted in their frequency. On the one hand, it is true that the ratio of turns to the set of all nonlevel maneuvers ($82/174 = 0.47$) is in the same direction as the ratio of utility scores in Table 1 ($1/[\text{mean of } 4] = 0.25$), both showing vertical to be strongly preferred over lateral, as predicted in the first hypothesis. On the other hand, though, contrasted against the separate categories of vertical maneuvers, turns showed greater utility than predicted. One possibility is that the penalties that the EU model imposed on turns that are not imposed on any of the four vertical maneuvers were too harsh in reality. A second possibility is that turns have another utility factor, not considered in the model; that they do not affect airspeed (and hence require an airspeed adjustment) in the manner of any of the forms of

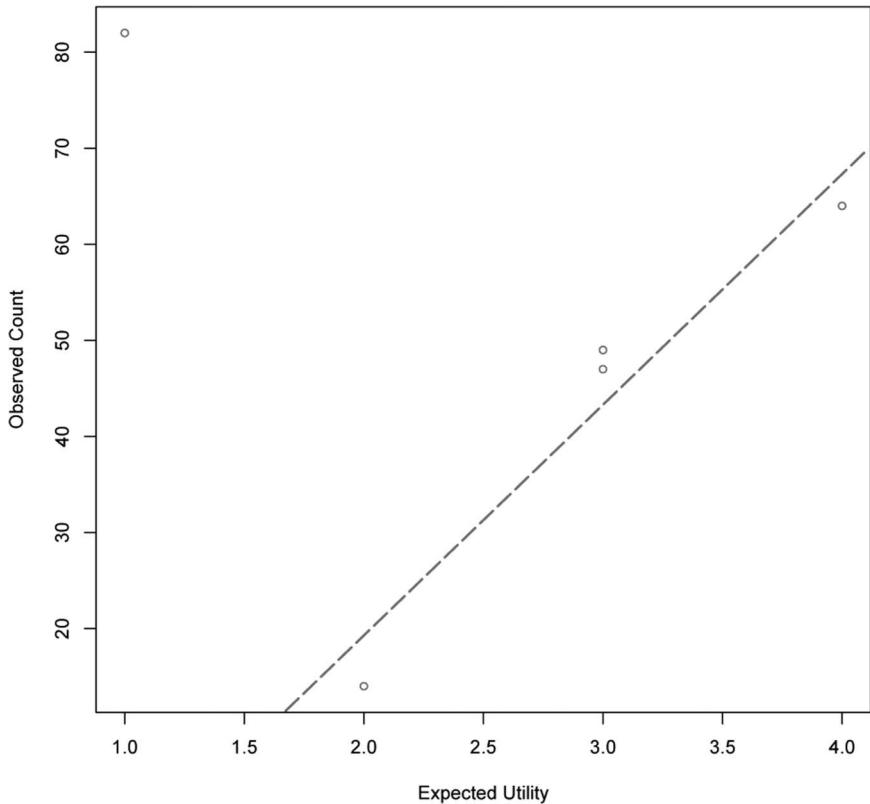


FIGURE 3 A scatterplot of expected utilities of different maneuvers versus the number of observed conflict resolution maneuvers. The regression line is based only on the four vertical maneuver classes, exhibiting nearly perfect fit (R^2 adjusted = 0.91) between the predictions and observations.

vertical maneuvers. It is difficult at this point to determine the extent to which one or the other of these might be responsible.

The results can be easily and plausibly explained also in terms of safe and sound ATC practices. Good controllers are quite aware of and indeed conscientious about the efficiency and economy of flights under their control. Therefore, climbing aircraft above their desired altitudes increases fuel burn, whereas descending an aircraft exploits gravity and does not significantly decrease efficiency, especially if the additional distance flown at the lower altitude (which is less optimal in terms of fuel burn) is short. Temporarily leveling off an aircraft already in descent makes little difference in terms of efficiency. There is also evidence of controllers' preference for vertical maneuvers, which provide for quicker

resolution to conflicts than lateral turns. Moreover, vertical maneuvers, descents in particular, might conform to aircrafts' planned trajectories, relieving controllers from the attention- and memory-intensive task of returning flights to their original courses after conflicts have been resolved (cf. Rantanen & Nunes, 2005).

In conclusion, there indeed appear to be regular patterns of conflict resolution maneuvers, emerging from reasonably large samples from five different ARTCCs in the United States. Although these patterns are not surprising in that they can be explained by good ATC practices, it is noteworthy that they emerged from objective data through objective analyses. To our knowledge, this is the first time analyses like these have been applied to data like these. However, we also have to acknowledge that this research was data-limited, both by quality (no voice data, no information about other traffic, some corrupted data) and quantity (a few hundred cases is not much when compared to the variability in conflict geometries and possible maneuver options). Nevertheless, we were able to develop and test methodologies that could be easily reapplied on bigger and better data sets, perhaps in longitudinal studies. Finally, we would like to make a case for continual analysis of operational data, which can reveal patterns and behaviors that could never emerge in simulated laboratory experiments. Routine access to data such as reported here is crucial for the research community to keep up with and contribute to the development of the NextGen systems. Objective evaluations of tactical ATC in general and conflict resolution practices in particular, combined with controllers' subjective assessment of NextGen technologies, would provide for a complete picture of the effects of these ATC and ATM modernization efforts.

ACKNOWLEDGMENTS

The original data were obtained through Grant OGC No. 20070271, addressing the Alert Problem in Air Traffic Control Facilities, from the FAA to New Mexico State University. The authors wish to thank Kenneth Allendoerfer, Ken Leiden, Stephen Rice, Krisstal Clayton, and Jill Kamienski for their contributions.

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